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A thermodynamic description of the B–Co system: modeling and experiment

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The B–Co system is investigated via thermodynamic modeling and experiments. In the modeling section, all of the experimental phase diagram and thermodynamic data available from the literature were critically reviewed and assessed by using thermodynamic models for the Gibbs energies of individual phases. In the experimental section, 6 key alloys were prepared by arc melting Co slug and B pieces and annealing at 900 °C for 9 days. Water-quenched samples were analysed by using X-ray diffraction, optical microscopy, electron probe microanalysis and differential thermal analysis techniques. The measured invariant reaction temperatures are: 1136 ± 2 °C for Liquid (L) $\leftrightarrow$ (αCo) + Co_3B, 1157 ± 2 °C for L + $Co_2B \leftrightarrow Co_3B$, 1263 ± 2 °C for L $\leftrightarrow Co_2B$ + CoB, and 1358 ± 2 °C for L $\leftrightarrow$ CoB + B. A consistent thermodynamic data set for the B–Co system is finally obtained by considering the present experimental results and reliable literature data. Comparisons between the calculated and measured phase diagram and thermodynamic quantities show that all of the accurate experimental information is satisfactorily accounted for by the thermodynamic description.

Keywords: Phase diagram B–Co; X-ray diffraction; Thermal analysis; Electron probe microanalysis; Thermodynamic calculations

Thermodynamische Beschreibung des Systems B–Co durch Modellrechnung und Experimente

Das System B–Co wurde mittels experimenteller Messungen und thermodynamischer Modellierung untersucht. Alle in der Literatur verfügbaren experimentellen thermodynamischen und Phasendiagramm-Daten wurden kritisch durchgesehen und mittels thermodynamischer Modelle für die freie Enthalpie bewertet. Danach wurden 6 charakteristische Legierungen aus Co und B im Lichtbogen erschmolzen und bei 900 °C 9 Tage lang geglüht. Die in Wasser abgeschreckten Proben wurden mit Röntgenbeugung, Licht- und Elektronenmikroskopie sowie Differenzialthermoanalyse untersucht. Es wurden folgende invarianten Temperaturen beobachtet: 1136 ± 2 °C für Schmelze (L) $\leftrightarrow$ αCo + Co_3B, 1157 ± 2 °C für L + $Co_2B \leftrightarrow Co_3B$, 1263 ± 2 °C für L $\leftrightarrow Co_2B + Co_3B$, und 1358 ± 2 °C für L $\leftrightarrow$ CoB + B. Mit diesen Ergebnissen sowie als verlässlich befundenen Literaturdaten wurde ein konsistenter Satz thermodynamischer Daten für das System B–Co erhalten. Der Vergleich von berechneten und beobachteten Phasendiagrammen und thermodynamischen Daten zeigt, dass sämtliche experimentelle Informationen durch diese thermodynamische Modellierung zufriedenstellend beschrieben werden.

1 Introduction

Alloys of the 3d transition metals (Co, Ni, Fe, Cr, and Mn) with B are reported to possess high hardness and resistance to wear in severe environments and thus ensure widespread industrial use [60Sey]. In addition, the rapidly quenched melts of Co with metalloids, such as B and C, are widely used in the production of amorphous materials, having high magnetic permeability and zero magnetostriction [81Bas, 86Lah, 99Her]. In order to improve the properties of these materials and design experiments most efficiently, information on phase equilibria and thermodynamic properties is needed. For example, the thermodynamic quantities for liquid alloys are critical for the prediction of glass-forming regions in these alloys. The phase relations and thermodynamic properties in the B–Co system have been the subject of extensive investigations since the early work by [38Kös]. However, the data reported in these studies show noticeable discrepancies.

The B–Co system has been previously modeled by three groups of researchers [84Kau, 88Lia, 99Mik], and all of them used two-term lattice stabilities for the pure elements. Today, more accurate lattice stabilities for pure elements are available [91Din]. In the modeling by [84Kau], limited experimental data were used. Consequently, their calculations show large discrepancies from the experimental phase diagram and thermodynamic data. The thermodynamic evaluation by [88Lia] was also considered to be unreliable since inaccurate enthalpy of mixing values [75Esi] were used in their optimization. The enthalpy of mixing data of [75Esi] were corrected later by the same authors [80Esi]. [99Mik] recently modeled the system and successfully reproduced most of the experimental data. But as many as 16 parameters were used to describe the thermodynamic property of the liquid phase in the modeling by [99Mik].

A thorough reassessment of the B–Co system is necessary in order to provide a reliable basis for thermodynamic extrapolations and calculations in related ternary and higher-order systems. The purpose of the present work is to provide a self-consistent set of thermodynamic parameters for the B–Co system by means of thermodynamic modeling coupled with key experiments.

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2 Evaluation of experimental data from the literature

2.1 Phase diagram data

[66Sch] is responsible for the major contributions to the understanding of the B–Co phase diagram. By means of optical microscopy, X-ray diffraction (XRD), and differential thermal analysis (DTA) techniques, these authors constructed the phase diagram from 20 at.% B to pure B in the temperature range 1000–1600 °C. In agreement with [59Kol], [66Sch] observed the intermediate phases Co_3B [58Run, 59Aro, 59Fru, 80Gia], Co_2B [07Jas, 33Bju, 59Aro, 63Sta, 72Hav, 80Rog], and CoB [07Jas, 33Bju, 63Sta, 78Pra, 80Rog]. In the phase diagram presented by [66Sch], the invariant reaction temperatures were taken to be the average values of peak onsets of DTA curves on heating and cooling. Since the supercooling phenomenon often accompanies solidification, the temperatures for invariant reactions are evaluated as the peak onset in the heating curves [98Du, 01Mic]. Consequently, the invariant reaction temperatures published by [66Sch] are estimated to be lower than the equilibrium temperatures. Table 1 shows both the invariant reaction temperatures reported by [66Sch] and assessed by the present authors considering the peak onsets of [66Sch] on heating only. In the work due to [66Sch], the liquidus lines and congruent melting temperatures for Co_2B and CoB are mainly based on the thermal effects in the heating curves. These data are considered to be reliable and, therefore, utilized in the present modeling.

The Co–Co_2B partial system was reinvestigated by [76Omo1], who employed XRD, DTA, and microscopic examination methods. The eutectoid decomposition temperature of Co_3B to (αCo) and Co_2B was determined to be about 845 °C by [76Omo1] using electromotive force (emf) measurement of the cells:<Pt> O_2 in air |$ZrO_2 \cdot CaO$| Co–Co_3B (or Co_2B), B_2O_3<Pt>. The existence of such a eutectoid decomposition was confirmed when [79Kuz] observed the three-phase equilibrium (αCo), Co_2B and $Ta_2Co_{21}B_6$ at 800 °C for the B–Co–Ta ternary system. The results from [76Omo1] confirm the general features of the Co–Co_2B partial system established earlier by [66Sch], although the former used starting materials with less purity [76Omo1]. As a consequence, the experimental data of [76Omo1] are used in the present optimization, but a low weight factor was applied to them.

Table 1. Comparison between the calculated and measured invariant reaction temperatures (°C) in the B–Co system.

Equilibrium	Measured			Calculated
	[66Sch][a]	[66Sch][b]	This work	This work
L ↔ (αCo) + Co_3B	1110 ± 5	1138 ± 5	1136 ± 2	1133
L + Co_2B ↔ Co_3B	1125 ± 5	1153 ± 5	1157 ± 2	1158
L ↔ Co_2B + CoB	1250 ±5	1261 ± 5	1263 ± 2	1271
L ↔ CoB + B	1350 ± 5	1355 ± 5	1358 ± 2	1353
L ↔ Co_2B	1280 ± 5	1280 ± 5	≈ 1285	1276
L ↔ CoB	1460 ± 5	1460 ± 5	> 1444	1462

[a] Temperatures as given in the phase diagram by [66Sch].
[b] Assessed by the present authors using the original DTA data on heating published by [66Sch]. The temperatures for invariant reactions are evaluated as the peak onsets in the heating curves.

Based on XRD and metallographic observations, [60Kol] obtained the phase diagram for the partial Co–Co_2B system. The liquidus temperatures measured by [60Kol] are lower by 20 to 100 °C than those from [66Sch, 76Omo1]. Consequently, the data of [60Kol] are excluded from the optimization.

[38Kös] constructed a partial phase diagram from Co to Co_2B by means of DTA and XRD methods. [40Chi] reinvestigated this partial system using thermal analysis, metallographic observations, XRD, and hardness measurements. Both groups of researchers failed to find the compound Co_3B and reported a eutectic reaction: liquid L ↔ (αCo) + Co_2B at about 1100 °C. This eutectic reaction was found to be metastable in the work of [66Sch]. The data associated with this metastable eutectic reaction are not utilized in the optimization, but they are used to compare with the calculation in metastable regions. The Co-rich liquidus temperatures reported by [38Kös, 40Chi] are the only literature data in this region. These liquidus temperatures are employed in the optimization but applied with a relatively low weight factor.

The solubility of B in Co at 1000 °C was determined to be 0.2 at.% B by [73Ler] using XRD and resistivity measurements. This result coincides well with that from [60Kol], who reported that the solubility of B in Co at 1095 °C is less than 0.16 at.% B by analysing the microstructures of the annealed samples. The solubility value of [73Ler] is retained in the present optimization.

2.2 Thermodynamic data

By means of high-temperature calorimetry, [75Esi] measured the enthalpy of mixing of the liquid at 1840 K. The starting materials they used were not of high purities (98 wt.% B and 99 wt.% Co). Later, the same group of authors [80Esi] recalculated their previous results [75Esi] by using the correct value for the enthalpy of melting of B. The corrected data [80Esi] are not used in the present optimization. However, they are compared with the calculated results in order to check the final modeling. [90Vit] measured the partial enthalpies of formation for B and Co at 1900 K by using a high-temperature isoperibolic calorimeter. These data were adopted by [95Wit], who critically evaluated the thermodynamic data of binary and ternary melts of the 3d transition metals (Cr, Mn, Fe, Co, and Ni) with B. Consequently, the experimental values from [90Vit] are used in the present modeling.

Using an adiabatic mixing calorimeter, [77Sid] measured the enthalpy increments ($H_{T,K} - H_{298\ K}$) of CoB in the temperature range 400–2000 K. [77Sid] also determined the heat capacity (C_p) of CoB at 300 K by means of an adiabatic calorimeter. Since [77Sid] employed high-purity materials (99.98 wt.%Co and 99.23 wt.%B), these heat capacity and enthalpy increment values were included in the present optimization.

[76Omo2] measured the enthalpies of formation ($\Delta_f H$) for Co_3B and Co_2B by means of an emf technique. Following the same method, [89Koy] obtained $\Delta_f H$ values of Co_3B, Co_2B, and CoB. Both [79Kub] and [82Sat] measured $\Delta_f H$ for Co_2B and CoB by using calorimetry. Within the estimated experimental errors, the values published by

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[89Koy] agree reasonably with those of [82Sat]. In the present modeling, the data from [82Sat, 89Koy] are utilized. Using a low-temperature calorimetry, [98Sir, 99Sir] measured the C_p values of Co_3B, Co_2B, and CoB in the temperature range 5–300 K. The entropies at 298.15 K derived from C_p values are included in the optimization.

Since a set of thermodynamic parameters for the whole B–Co system could be obtained by using the experimental phase diagram and thermodynamic data indicated in the preceding sections, it should be of considerable interest to compare the calculated thermodynamic properties with the corresponding experimental values, which are not utilized in the modeling. Activities of B in the B–Co melt at 1200 and 1300 °C were determined by [89Yuk] using an emf method. The activity data from [89Yuk] are used for such a comparison.

3 Key experimental investigation

3.1 Experimental procedure

Prior to the present experiment, a preliminary thermodynamic modeling for the B–Co system is performed by using the experimental data selected in Section 2. The computed B–Co phase diagram is used to guide the choice of the compositions of the decisive alloys. Another reason to conduct key experiments is that even the most detailed measurements [66Sch] failed to obtain accurate data on invariant reaction temperatures, as pointed out in the preceding section.

The starting materials used for the present investigation are 99.95 wt.% purity Co slug and 99.5 wt.% purity B pieces (both from Johnson Matthey Company, MA 01835, USA). Alloys with a weight of about 1 g were prepared by melting Co slug and B pieces with an arc furnace in an ambient atmosphere of 99.998 % purity Ar. The vacuum inside this arc furnace was pumped below ≈ 5 Pa before the purge of Ar. The as-cast alloys were sealed in evacuated silica tubes under vacuum (1 Pa), annealed at 1000 °C for 9 days, and then water-quenched. The compositions of the alloys are listed in Table 2. Chemical analysis for the alloys was conducted since the weight losses during arc-melting were generally about 2 wt.%. The elemental concentrations were determined with a PQ2 Turbo Plus ICP-MS (inductively-coupled plasma mass spectrometer) instrument (VG PlasmaQuad, Fison, Loughborough, Leicestershire, UK).

The phase identification in the quenched alloys was performed by means of XRD (PW1729, X-ray generator, Philips) using Cu Kα_1 radiation with Si as an internal standard. All of the quenched alloys were ground, polished and then analysed by optical microscopy. The alloys were also investigated by electron probe microanalysis (EPMA) (CAMECA SX 50, France) in order to confirm optical microscopy observations and XRD analyses. Pure Co and B, which were subjected to the same experimental procedure as that employed for the alloys, were used as standards. Conventional matrix correction, which treats the deviation from linearity by including the effects of atomic number (Z), absorption (A) and fluorescence (F), was utilized to calculate the compositions from the measured X-ray intensities for B Kα and Co Kα with an accelerating voltage of 15 kV and a beam current of 20 nA.

DTA (DTA 701L, Bähr Thermoanalyse GmbH, Germany) measurements of the alloys were carried out in Al_2O_3 crucibles under a flow of pure Ar. The measurements were conducted between room temperature and 1500 °C with a heating and cooling rate of 5 K/min. The temperature was measured with Pt–Pt/Rh thermocouples and calibrated to the melting temperatures of Al (660.3 °C), Ag (961.8 °C), Au (1064.2 °C), Si (1413.9 °C), Ni (1455.2 °C), and Co (1494.9 °C) [90Tho, 91Din]. The accuracy of the temperature measurement was estimated to be ± 2 °C. The invariant reaction temperatures were determined from the onset of the thermal effects during the heating step, and the offset temperatures of the last thermal effect on heating were taken for the liquidus.

3.2 Experimental results

The invariant reaction temperatures derived from the DTA signals of the annealed alloys are included in Table 1. Table 2 summarizes the phases identified by XRD and optical microscopy, phase compositions obtained by EPMA as well as the transition temperatures resulting from DTA measurements. In these measurements, which were found to be consistently reproducible, the general features of the B–Co phase diagram established by [66Sch] are confirmed, but for the three-phase equilibrium temperatures noticeable deviations are observed, as shown in Table 1. Most prominently, Co-rich eutectic (L ↔ (αCo) + Co_3B) and peritectic (L + Co_2B ↔ Co_3B) reaction temperatures are higher by 26 and 28 °C, respectively, than those given by [66Sch]. As indicated before, the present authors have reassessed the DTA data of [66Sch] by taking the peak onset in the heating curve as the invariant reaction temperature. It is worth noting that these assessed temperatures agree well with the presently measured ones. XRD data confirmed all established B–Co intermetallic phases and their crystal structures. The observed lattice parameters match closely with those re-

Table 2. Summary of the phases identified by XRD and metallography, phase compositions by EPMA as well as DTA peaks for the B–Co alloys annealed at 1000 °C for 9 days.

Composition (at.% B)		Phase	Phase composition (at.% B)	DTA signal[b] (°C)
Nominal	Analysis			
9	7.2	(αCo) + Co_3B	(αCo) : 0.28; Co_3B : 26.3	1136, 1459
25	24.5	Co_3B + (αCo)[a]	Co_3B : 26.4; (αCo)[a] : 0.25	1137[c], 1157, 1204
33.3	33.6	Co_2B + CoB[a]	Co_2B : 36; CoB : 51.7	1265, ≈1285
40	38.5	Co_2B + CoB	Co_2B : 36.3; CoB : 51.8	1263, 1300
50	49.4	CoB	CoB : 52	1263[c], 1358[c], ≈1444
63	62.6	CoB + (B)	CoB : 51.5; (B) : 99.4	1358, 1417

[a] Means trace amount of the phase.
[b] Obtained on heating with a heating rate of 5K/min.
[c] Very weak peaks.

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ported by [91Vil] and are thus not presented here. EPMA measurements confirm the phase compositions reported in the literature, and the measured solubility for B in Co (0.28 at.% B at 1000 °C) is in good agreement with that (0.2 at.% B at 1000 °C) from [73Ler].

Table 3. Summary of the thermodynamic parameters of the B–Co system.[a]

Liquid: $(B, Co)_1$

$^0L^{L}_{B,Co} = -162\,096.49 + 48.36349 \cdot T$

$^1L^{L}_{B,Co} = -2\,542.75 + 4.71172 \cdot T$

$^2L^{L}_{B,Co} = 32\,123.74$

fcc-(αCo):[b] $(B, Co)_1$

$^0L^{(\alpha Co)}_{B,Co} = -18\,734.7$

hcp-(εCo):[b] $(B, Co)_1$

$^0L^{(\varepsilon Co)}_{B,Co} = 0$

Co_3B: $Co_{0.75}B_{0.25}$

$^0G^{Co_3B}_m - 0.75 \cdot {}^0G^{hcp}_{Co} - 0.25 \cdot {}^0G^{rhombohedral}_B =$
$= -21\,131.79 + 3.19280 \cdot T$

Co_2B: $Co_{2/3}B_{1/3}$

$^0G^{Co_2B}_m - (2/3) \cdot {}^0G^{hcp}_{Co} - (1/3) \cdot {}^0G^{rhombohedral}_B =$
$= -28\,563.67 + 5.07674 \cdot T$

CoB: $Co_{0.5}B_{0.5}$

$^0G^{CoB}_m - 0.5 \cdot H^{SER}_B - 0.5 \cdot H^{SER}_B =$
$= -40\,616.6 + 134.82159 \cdot T - 21.3264 \cdot T \ln T$
$-0.0041898 \cdot T^2 + 297\,858.09 \cdot T^{-1}$

rhombohedron–(B): B

[a] In J/mol-atoms; temperature (T) in Kelvin. The Gibbs energies for Co in liquid, fcc, and hcp forms and those for B in liquid and rhombohedral structure are from the SGTE compilation [91Din]. The Gibbs energies for B in metastable fcc and hcp modifications are taken from [84Kau].

[b] The magnetic contribution to the Gibbs energy is described by Hillert-Jarl-Inden model [78Hil].

4 Thermodynamic model

The Gibbs energy function $^0G^{\Phi}_i(T) = G^{\Phi}_i(T) - H^{SER}_i$ for the element i (i = B, Co) in the phase Φ is expressed by an equation of the form:

$$^0G^{\Phi}_i(T) = a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1} + f \cdot T^3 + g \cdot T^7 + h \cdot T^{-9} \quad (1)$$

where H^{SER}_i is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, and T is the absolute temperature. The last two terms in Eq. (1) are used only outside the ranges of stability [87And], the term $g \cdot T^7$ for a liquid below the melting point and $h \cdot T^{-9}$ for solid phases above the melting point.

The Gibbs energy of the liquid is described by the Redlich–Kister polynomial [48Red]:

$$G^L_m - H^{SER} = x_{Co} \cdot {}^0G^L_{Co} + x_B \cdot {}^0G^L_B + R \times T(x_{Co} \cdot \ln x_{Co} + x_B \cdot \ln x_B) + x_{Co} \times x_B[a_0 + b_0 \cdot T + (x_B - x_{Co})(a_1 + b_1 \cdot T) + (x_B - x_{Co})^2 \times (a_2 + b_2 \cdot T) + \ldots] \quad (2)$$

in which H^{SER} is the abbreviation of $x_{Co} \cdot H^{SER}_{Co} + x_B \cdot H^{SER}_B$, R is the gas constant, and x_{Co} and x_B are the mole fractions of Co and B, respectively. The coefficients a_j and b_j (j = 0, 1, 2) are the parameters to be optimized.

The compounds Co_3B and Co_2B are modeled as stoichiometric ones. The Gibbs energy of Co_3B per mol-formula is given by the following expression:

$$^0G^{Co_3B}_m - 3 \cdot H^{SER}_{Co} - H^{SER}_B = A + B \cdot T + 3 \cdot {}^0G^{hcp-Co}_{Co} + {}^0G^{rhombohedral}_B \quad (3)$$

in which A and B are the parameters to be evaluated in the course of optimization. The parameter $^0G^{rhombohedral}_B$ is the Gibbs energy of pure B in its stable rhombohedral structure. An analogous equation can be written for the Gibbs energy of Co_2B.

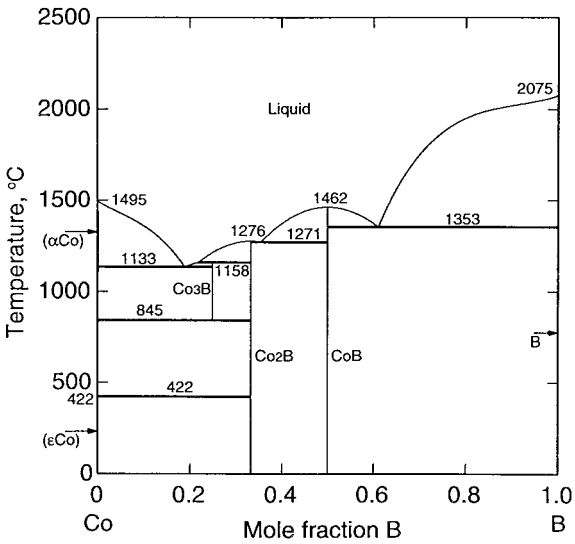

Fig. 1. Calculated B–Co phase diagram according to the present modeling.

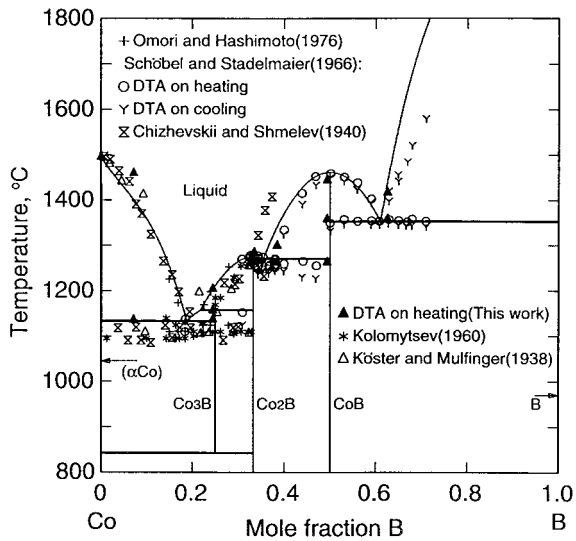

Fig. 2. Calculated B–Co phase diagram with experimental data from the present work and literature [38Kös, 40Chi, 60Kol, 66Sch, 76Omo1].

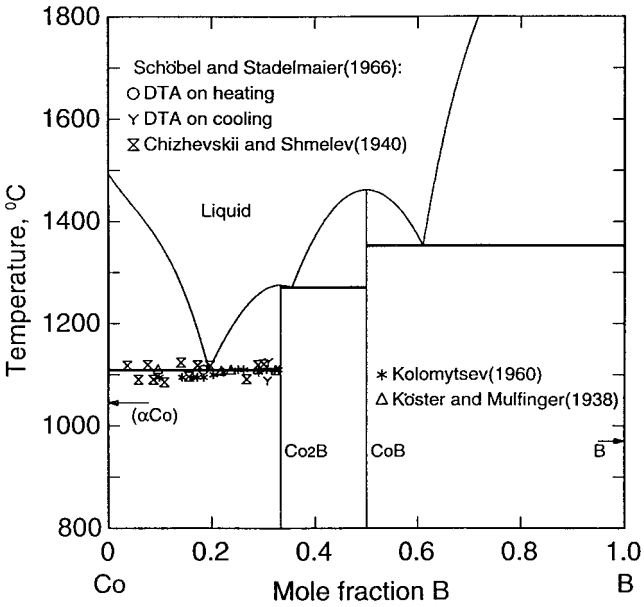

Fig. 3. Calculated metastable B–Co phase diagram with the experimental data on the metastable eutectic reaction L ↔ (αCo) + Co_2B [38Kös, 40Chi, 60Kol, 66Sch]. The stable compound Co_3B is suspended in the calculation.

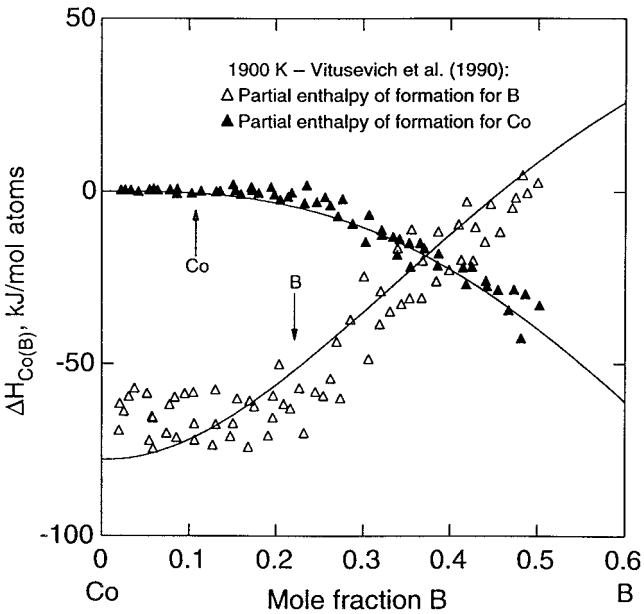

Fig. 5. Calculated partial enthalpies of mixing for B and Co at 1900 K, compared with the experimental data [90Vit]. The reference states are solid B and liquid Co.

Since there are experimental data on C_p of CoB in a wide temperature range [77Sid], it is preferable to express its Gibbs energies relative to the SER state. The Gibbs energy of CoB per mol-formula is expressed as:

$$G_m^{CoB} - H_{Co}^{SER} - H_B^{SER} = a + b \cdot T + c \cdot T \cdot \ln T + d \cdot T^2 + e \cdot T^{-1} \quad (4)$$

The coefficients c, d, and e can be obtained directly from the C_p expression.

5 Results and discussion

The model parameters were evaluated using the computer-operated optimization program PARROT [85Sun], which works by minimizing the square sum of the differences between measured and calculated values. The step-by-step optimization procedure carefully described by [97Du] was utilized in the present assessment. The experimental data selected from the literature as well as the present experimental results are employed in the optimization.

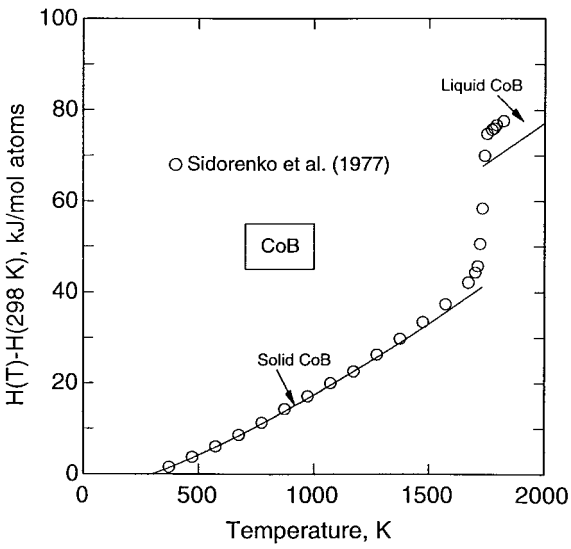

Fig. 4. Calculated $H_T - H_{298\,K}$ for the alloy $Co_{0.5}B_{0.5}$ with the measured data of [77Sid].

[77Sid] derived the C_p expression, $(20 + 0.011 \cdot T - 4.4 \cdot 10^5 \cdot T^{-2})$ J/(mol-formula · K), of CoB from their measured enthalpy increment data. The coefficients c, d, and e in Eq. (4) are related to this expression by the following equations: $c = -2 \cdot 20$, $d = -0.011$, and $e = 4.4 \cdot 10^5$. When c, d, and e have been estimated, the coefficient a in Eq. (4) can be evaluated from the $\Delta_f H$ value [82Sat], and the coefficient b from the measured phase diagram data. The obtained coefficients for CoB were subjected to further optimization. For Co_3B and Co_2B, the coefficients A and B in Eq. (3) are adjusted to reproduce the phase diagram and/or $\Delta_f H$ data concerning them. For the (αCo) terminal solid solution, a constant regular parameter was evaluated to fit the solubility of B in αCo [73Ler]. In the case of (εCo) solid solution, an ideal solution model was used since there is virtually no solubility of B in εCo. B is treated as a pure element. For the liquid, a_0 and a_1 in Eq. (2) need to be introduced in order to make the excess enthalpy of B-rich liquid independent of that of Co-rich liquid. The well-established liquidus line in the wide temperature and composition ranges makes it possible to optimize the parameters b_0 and b_1. In the present work, it was found that the introduction of a further parameter, a_2, can improve the description for the liquid phase. The thermodynamic parameters, finally obtained in the present work, are listed in Table 3.

The complete B–Co phase diagram calculated using the present set of thermodynamic parameters is shown in Fig. 1. The Pandat [00Pan] and Winphad [00Win] programs are used to check if the computed phase diagram is a real stable one, and the calculated invariant reaction temperatures are compared with the experimental ones in Table 1.

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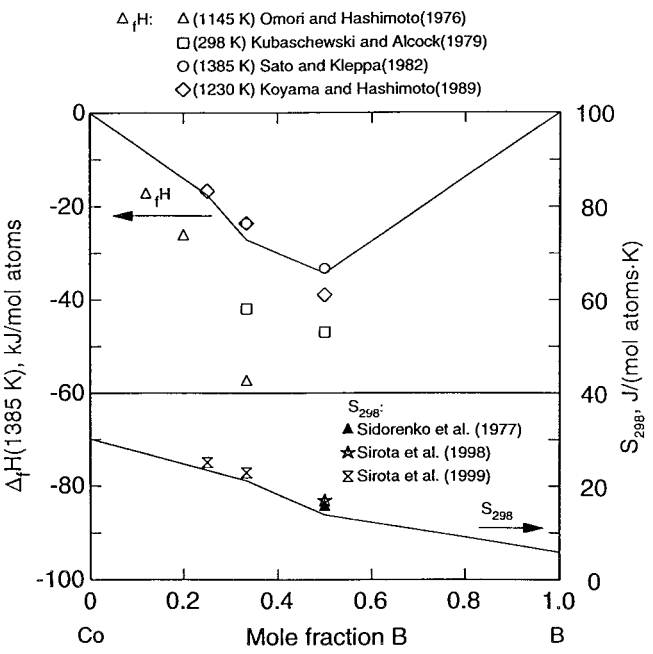

Fig. 6. Calculated enthalpy of formation and entropy, compared with the experimental data [76Omo2, 77Sid, 79Kub, 82Sat, 89Koy, 98Sir, 99Sir].

It can be seen in Table 1 that the present modeling can account for the reassessed invariant reaction temperatures of [66Sch] and those from the present work within 8 K.

A comparison of the computed B–Co phase diagram with the newly measured values is shown in Fig. 2, in addition to the literature data. The fit to the DTA data on heating from [66Sch, 76Omo1] and this work is excellent. As expected, the DTA data on cooling from [66Sch] are generally below the calculated curves. The measured Co-rich invariant reaction temperatures, which are below the calculated $L \leftrightarrow (\alpha Co) + Co_3B$ reaction temperature (Fig. 2), can be elucidated from Fig. 3 where the calculated metastable $L \leftrightarrow (\alpha Co) + Co_2B$ eutectic reaction agrees well with the corresponding experimental data [38Kos, 40Chi, 60Kol, 66Sch]. In fact, both [38Kos] and [40Chi] only observed the phases (αCo) and Co_2B in the Co-rich part of the system, and their data should correspond to the metastable eutectic equilibrium $L \leftrightarrow (\alpha Co) + Co_2B$. This metastable eutectic reaction was also observed by [66Sch].

In Fig. 4, the calculated enthalpy increment for the alloy $Co_{0.5}B_{0.5}$ is compared with the experimental values [77Sid]. The agreement is good within estimated experimental uncertainties, except for some deviation at high temperatures. That deviation is probably due to the shift in the compositions of the investigated alloys in the liquid state, as mentioned by [77Sid]. Figure 5 shows a comparison of the calculated partial enthalpies of formation for B and Co at 1900 K with the experimental data of [90Vit]. The calculation can satisfactorily account for the scattering experimental values. A further check on the reliability of the thermodynamic modeling is provided by Fig. 6, where the calculated enthalpies of formation and entropies for the compounds Co_3B, Co_2B, and CoB are compared with the experimental data [76Omo2, 77Sid, 79Kub, 82Sat, 89Koy, 98Sir, 99Sir]. It is demonstrated that the reliable experimental data [77Sid, 82Sat, 89Koy, 98Sir, 99Sir] are well described by the thermodynamic calculation.

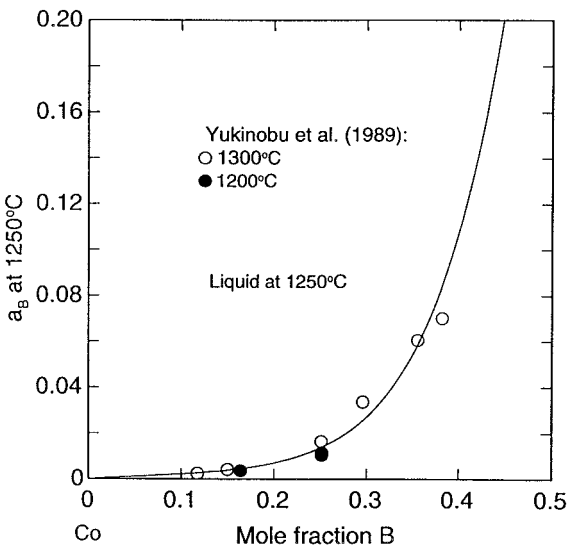

Fig. 7. Model-predicted activity of B in the liquid with the experimental values from [89Yuk]. The reference state is solid B.

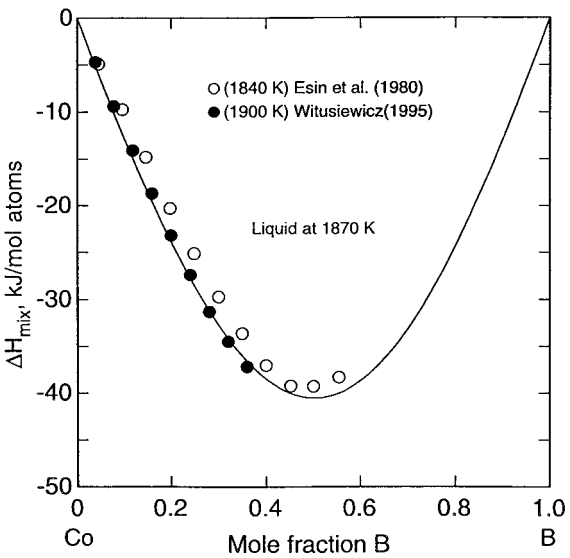

Fig. 8. Model-predicted enthalpy of mixing for the liquid at 1870 K with the experimental values from [80Esi] as well as the evaluated values of [95Wit]. The reference states are liquid B and liquid Co.

In this paragraph, we compare the thermodynamic quantities predicted by using the thermodynamic parameters with two pieces of experimental information, which are not used in the modeling. Figure 7 compares the predicted activity of B in the liquid at 1250 °C with the experimental values of [89Yuk], showing an excellent agreement. A comparison of the model-calculated enthalpy of mixing with the experimental data [80Esi] and the assessed values [95Wit] is made in Fig. 8 and shows also a good agreement.

The thermodynamic parameters of the B–Co system obtained in the present work have been used to calculate the phase relationships in the B–Co–Cr and B–Co–Fe systems successfully [02Hsi].

6 Conclusions

1. Using 6 decisive alloys, which are analysed with XRD, DTA, optical microscopy and EPMA techniques, we have checked the B–Co phase diagram. All of the experimen-

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tal data available for the B–Co system have been critically evaluated. In particular, the original data of [66Sch] are reassessed in detail. The reassssed data are in excellent agreement with the present experimental results.

2. An optimal thermodynamic data set for the B–Co system was obtained by considering the present experimental results as well as critically evaluated literature data. The comprehensive comparison shows that the calculated phase diagram and thermodynamic properties are in very good agreement with the experimental information. The formerly measured metastable eutectic reaction data ($L \leftrightarrow (\alpha Co) + Co_2B$) can be well described by a metastable phase diagram in which the stable compound Co_3B is suspended from the calculation.

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